Software Engineering & Project

Snapshot week 8 - BLOCK UG-16

Maptek, Block Model Compression Algorithm

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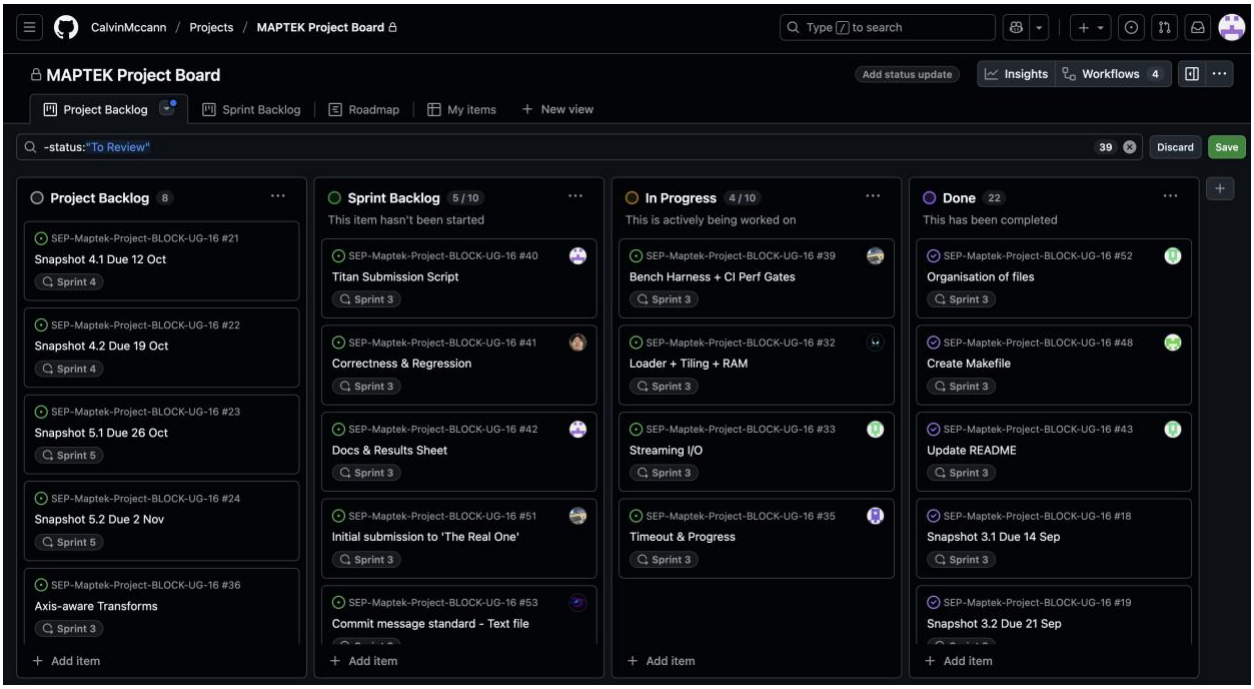
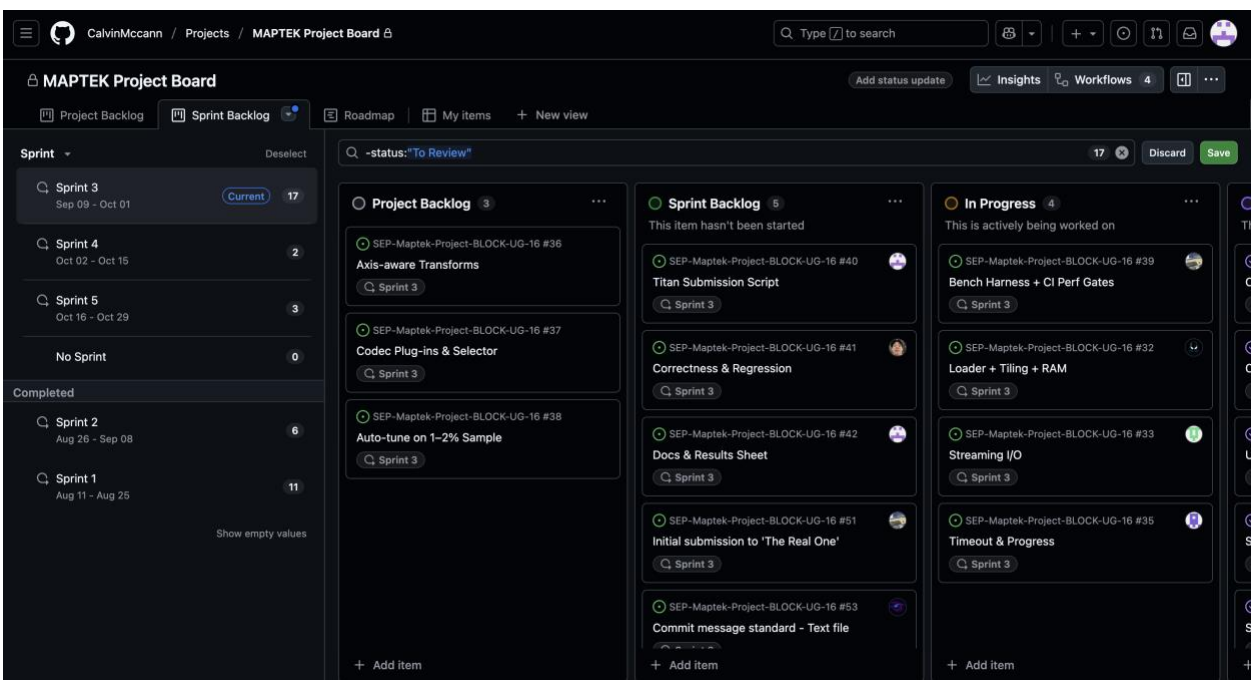
Product Backlog and Task Board

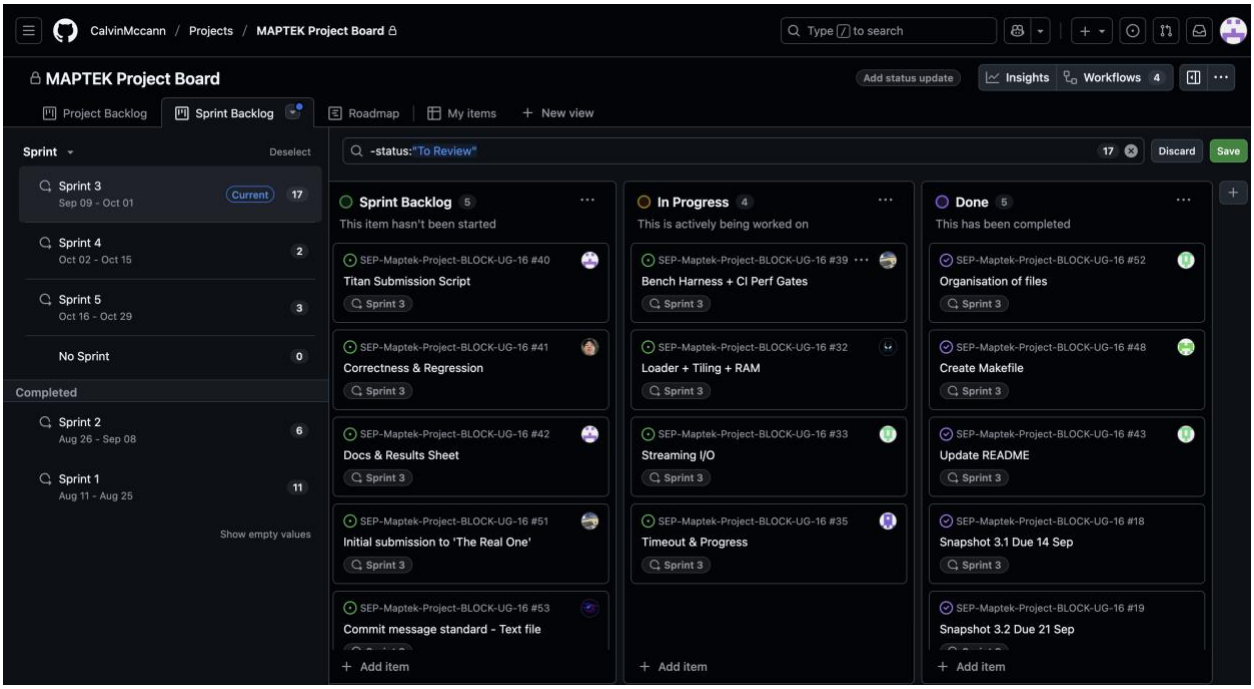
Product Backlog	
Task	**Description**
Axis-aware Transforms	delta/RLE per axis Palette/dictionary per tile
Snapshot 4.1	Create snapshot 4.2
Snapshot 4.2	Create snapshot 4.2
Snapshot 5.1	Create snapshot 5.1
Snapshot 5.2	Create snapshot 5.2
Correctness and Regression	Do: run current code on intro and fast competitions; confirm outputs and acceptance still pass. Done when: both older comps pass their acceptance checks.
Docs & Results Sheet	Shared results sheet (commit, params, ratio, MB/s, RAM, runtime)
Final Presentation	Initial planning for final presentation



Sprint Backlog and User Stories

Sprint Backlog	
Task	**Description**
Docs & results sheet	Shared results sheet (commit, params, ratio, MB/s, RAM, runtime)
Correctness & regression	Do: run current code on Intro and Fast competitions; confirm outputs and acceptance still pass. Done when: both older comps pass their acceptance checks.
Initial submission	Initial working submission to 'The Real One'
Commit Message Standard	Standardisation and documentation of expected commit messages.
Loader + Tiling + RAM	read data in tiles/chunks; reuse buffers; no full dataset in memory; add bounds/stride asserts.
Streaming I/O	prototype buffered read, mmap, and prefetch thread; time each; keep the fastest path under 8GB.
Timeout & progress	global wall-clock timer; at 80m either exit cleanly or switch to "fast mode"; write summary stats before exit.





User Stories (Sprint 3)

For a user of the block model compression algorithm, the algorithm needs to be optimised for speed, rigorously tested, and supported by more organised workflows so that larger datasets can be managed efficiently, and results trusted. In this sprint, the team will focus on improving performance by refining existing working code and reducing inefficiencies, while also implementing more structured testing to validate correctness and reliability across a wider range of block model scenarios.

To ensure these improvements are sustainable, workflows and project structure have been made cleaner and more consistent, making collaboration smoother and reducing the likelihood of errors. Functional algorithms from previous sprints will be revisited and polished to improve clarity, maintainability, and performance. As the Product Owner (TA), I have reinforced the importance of balancing optimisation with stability and worked with the team to set realistic goals that keep progress aligned with project deadlines and quality standards.

Definition of Done

Optimisation of the working compression algorithm such that it performs well on ‘The Real One’ compression scenario and works faster and more optimally on previous challenges.

Summary of Changes

Several tasks were ticked off the sprint backlog this week, including the creation of a makefile. This simplifies the build process for our compression algorithm, saving time in getting it accepted on Titan. The file structure and organisation task was also completed, ensuring every part of the project is easily located.

Every member of the team is working away on their tasks, ensuring all the sprint backlog tasks are completed by the end of sprint 3 in two weeks. WE expect to have a well performing algorithm submitted to Titan before Sprint 3 ends, and students return to university.